AS Chemistry — Topic 1 Atomic Structure (Exam · Paper B)

CIE 9701 AS Chemistry · Paper B · 33 marks · about 40–45 minutes

Topic 1: particles and isotopes, mass spectrometry, electronic configuration / orbitals, and ionisation energy. Answer in English. Section A is Paper 1 style (1 mark each). Section B is written; show working and give the usual CIE reasons (nuclear charge, radius, shielding, spin-pair repulsion) where asked. A Periodic Table may be used.

Section	**Content**	**Marks**
A	Multiple choice	13
B1	Iron isotopes, A_r, configuration and particles	10
B2	Tellurium: sub-shell and successive ionisation energy	6
B3	Mass spectrum of hydrocarbon R: n(C) and a fragment	4
	Total	**33**

Section A — Multiple-choice questions (13 marks)

A1. [1]

Which statement about $^{131}_{53}I$ is correct?

A A negative ion of $^{131}_{53}I$ contains 53 neutrons and 52 electrons.
B A negative ion of $^{131}_{53}I$ contains 53 neutrons and 54 electrons.
C A negative ion of $^{131}_{53}I$ contains 78 neutrons and 52 electrons.
D A negative ion of $^{131}_{53}I$ contains 78 neutrons and 54 electrons.

A2. [1]

Technetium (Tc) is a second row transition element that does not occur naturally on Earth. One of its isotopes has 56 neutrons. What is the nucleon number of this isotope?

A 43 **B** 56 **C** 99 **D** 112

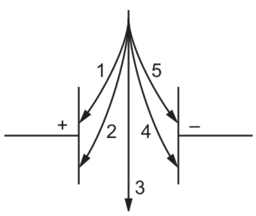
A3. [1]

In which row do the particles increase in size?

	smallest	⟶	largest
A	N	O	F
B	N^{3-}	O^{2-}	F^-
C	Na^+	Mg^{2+}	Al^{3+}
D	Na^+	Ne	F^-

A4. [1]

A helium ion contains two protons, two neutrons and one electron.
This helium ion and a proton are passed separately through a uniform electric field.
The particles are travelling at the same velocity.
Which arrow describes the path of each particle?



	helium ion	proton
A	1	2
B	2	1
C	4	5
D	5	4

A5. [1]

A sample of magnesium contains the isotopes ^{24}Mg, ^{25}Mg and ^{26}Mg only.
The percentage abundance of ^{25}Mg and ^{26}Mg is the same.
The relative atomic mass of magnesium in the sample is 24.3.
What is the percentage abundance of ^{24}Mg?

A 10% **B** 20% **C** 60% **D** 80%

A6. [1]

What is the electronic configuration for the particle $^{27}_{13}Al^{+}$?

A $1s^2\,2s^2\,2p^6\,3s^1\,3p^1$
B $1s^2\,2s^2\,2p^6\,3s^2$
C $1s^2\,2s^2\,2p^6\,3s^2\,3p^2$
D $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^7\,4s^1$

A7. [1]

Which statement about a 3p orbital is correct?

A It can hold a maximum of six electrons.
B It has the highest energy of the orbitals with principal quantum number 3.
C It is at a higher energy level than a 3s orbital but has the same shape.
D It is occupied by one electron in an isolated phosphorus atom.

A8. [1]

In which pair of species do both species have only one unpaired p electron?

A Ar^{+} and C^{-}
B B and Ti^{+}
C F and Ga
D Se^{-} and Si^{-}

A9. [1]

Why is the first ionisation energy of oxygen less than that of nitrogen?

A The nitrogen atom has its outer electron in a different subshell.
B The nuclear charge on the oxygen atom is greater than that on the nitrogen atom.
C The oxygen atom has a pair of electrons in one p orbital that repel one another.
D There is more shielding in an oxygen atom.

A10. [1]

The second ionisation energy of oxygen is greater than the second ionisation energy of fluorine.

Which factor explains this difference?

A The atomic radius of an oxygen atom is smaller than that of fluorine.
B The covalent bond in a fluorine molecule is weaker than the bond in an oxygen molecule.
C A spin-paired electron is removed from fluorine but **not** from oxygen.
D Fluorine has more electrons in total than oxygen. This causes a greater shielding of the nuclear attraction in fluorine.

A11. [1]

The first seven ionisation energies of an element between lithium and neon in the Periodic Table are shown.

1310 3390 5320 7450 11000 13300 71000 $kJ\,mol^{-1}$

What is the outer electronic configuration of the element?

A $2s^2$
B $2s^2\,2p^1$
C $2s^2\,2p^4$
D $2s^2\,2p^6$

A12. [1]

Which of these elements has the highest fifth ionisation energy?

A C **B** N **C** P **D** Si

A13. [1]

The first ionisation energy of potassium, K, is $418\,kJ\,mol^{-1}$. The first ionisation energy of strontium, Sr, is $548\,kJ\,mol^{-1}$.

Which statement helps to explain why Sr has a greater first ionisation energy than K?

A The charge on a Sr nucleus is greater than the charge on a K nucleus.
B The outer electron in a Sr atom experiences greater shielding than the outer electron in a K atom.
C The outer electron in a Sr atom experiences spin-pair repulsion.
D The outer electron in a Sr atom is further from the nucleus than the outer electron in a K atom.

Section B — Structured questions (20 marks)

B1 · Iron isotopes, relative atomic mass, configuration and particles (10 marks)

A sample of iron contains three different isotopes and has a relative atomic mass, A_r, of 55.8.

B1(a)(i) [2]

Define relative atomic mass.

B1(a)(ii) [2]

Table 2.1 shows the abundances of two of the isotopes present in the sample of iron.

isotope	relative isotopic mass	abundance / %
^{54}Fe	53.9	6.0
^{56}Fe	55.9	91.9

Table 2.1

Use Table 2.1 to calculate the relative isotopic mass of the third isotope of iron in the sample.
Show your working.

relative isotopic mass = ____________________

B1(b) [1]

Deduce the number of pairs of electrons in the 3d sub-shell in an iron(II) ion. number of pairs of electrons

____

B1(c) [1]

Sketch the shape of the lowest energy orbital in the shell with principal quantum number $n = 2$.

B1(d) [2]

Complete Table 2.2 to show information about particles in one ion of $^{56}Fe^{3+}$.

particle	number of particles present in one ion of $^{56}Fe^{3+}$
electrons	
	30

Table 2.2

B1(e) [2]

The atomic radius of iron is 1.26×10^{-10} m.
Suggest the change to the radius, if any, after an iron atom reacts to produce an Fe^{3+} ion.
Explain your answer.

B2 · Tellurium: sub-shell and successive ionisation energy (6 marks)

Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te. Its electronic configuration is [Kr] $4d^{10}\,5s^2\,5p^4$.

B2(a) [1]

Identify the sub-shell in an atom of Te that contains electrons with the lowest energy. sub-shell ____________________

B2(b) [1]

Construct an equation to represent the first ionisation energy of Te.

B2(c)(i) [2]

The radius of Te ions decreases after each successive ionisation.

State **two** factors that are responsible for the increase in the first six ionisation energies of Te.

B2(c)(ii) [2]

Sketch a graph in Fig. 1.1 to show the trend in the first **seven** ionisation energies of Te.

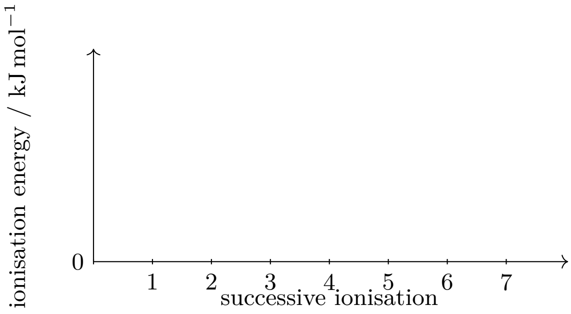


Fig. 1.1

B3 · Mass spectrum of hydrocarbon R (4 marks)

P, Q and R are three different hydrocarbon molecules.

The mass spectrum of hydrocarbon R is recorded. Information about the two peaks with m/e greater than 135 is shown in Fig. 6.1.

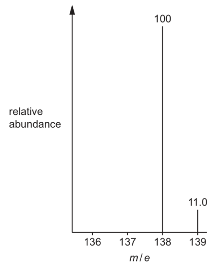


Fig. 6.1

B3(a) [2]

Use Fig. 6.1 to deduce the number of carbon atoms in a molecule of R.

Show your working.

number of carbon atoms = ____________________

B3(b) [1]

Use Fig. 6.1 and your answer to (a) to deduce the molecular formula of R.

B3(c) [1]

Suggest the molecular formula of the fragment of R with $m/e = 57$.